

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Seventh Semester B.Tech Degree Examination December 2022 (2019 scheme)

Course Code: CST413**Course Name: MACHINE LEARNING****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

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|----|--|-----|
| 1 | Explain a scenario in which classification is better than regression. | (3) |
| 2 | Identify the suitable learning method in each case and explain it | (3) |
| | a) Predict Stock Prices | |
| | b) Detect an email as spam or not | |
| 3 | Explain the general MLE method for estimating the parameters of a probability distribution | (3) |
| 4 | Compare linear regression model and logistic regression model | (3) |
| 5 | Construct a feed-forward network with four input nodes and three hidden nodes and three output nodes that has lateral inhibition structure in the output layer. | (3) |
| 6 | What is the significance of optimal separating hyperplane in SVM? | (3) |
| 7 | For the K-Means algorithm: Which parameters are usually specified by the user, and what does the algorithm estimate? | (3) |
| 8 | Is EM algorithm a supervised or an unsupervised learning method? Explain your answer. | (3) |
| 9 | What is ensemble learning? Can ensemble learning using linear classifiers learn classification of linearly non-separable sets? | (3) |
| 10 | Suppose there are three classifiers A,B and C. The (FPR, TPR) measures of the three classifiers are as follows – A (0, 1), B (1, 1) , C (1,0.5). Which can be considered as a perfect classifier? Justify your answer. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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|----|---|-----|
| 11 | a) You have a coin that you think is biased. you flip it 4 times and get the sequence HHHT. What is the maximum likelihood estimate for the probability of getting heads? | (4) |
|----|---|-----|

- b) The number of threes made during an NBA game is Poisson distributed. Last Saturday, the number of threes made were 14, 26, 25, and 13. Calculate the maximum likelihood estimate for the parameter (5)
- c) Suppose x is the number of trials to achieve the first success in a Bernoulli process with success probability p . Show that the maximum likelihood estimator of p is $\hat{p} = 1/x$. (5)

OR

- 12 a) Suppose $X_1, X_2, \dots, X_n$ are independent and identically distributed random variables with the common probability mass function (pmf): (4)
- $$p(x) = \theta(1 - \theta)^{x-1}, \text{ for } x=1,2,\dots \text{ and } 0 < \theta < 1$$
- Write down the likelihood function of θ .
- b) Find the maximum likelihood estimator (MLE) of θ . (7)
- c) Suppose you roll a 6-sided die 40 times and observe eight rolls of a 6. What is the maximum likelihood estimate of the probability of observing a 6? (3)

Module II

- 13 a) Consider the following data for the binary classification. Calculate the information gain for each attribute and draw the decision tree by selecting the best split (9)

Tid	Refund	Marital status	Taxable income	Class
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- b) What are the issues faced by decision tree-based classification algorithms? (5)

OR

- 14 a) Given the following data on a certain set of patients seen by a doctor. Can the doctor conclude that a person having chills, fever, mild headache and without running nose has flu? (Use Naive Bayes classification). (9)

Chills	Running nose	Headache	Fever	Has flu
Y	N	Mild	Y	N
Y	Y	No	N	Y
Y	N	strong	Y	Y
N	Y	mild	Y	Y
N	N	no	N	N
N	Y	strong	Y	Y
N	Y	strong	N	N
Y	Y	mild	Y	Y

- b) Obtain a linear regression for the data given in the table below assuming that y is the independent variable. (5)

x	55	60	65	70	80
y	52	54	56	58	62

Module III

- 15 a) What is a kernel in SVM? Why do we use kernels in SVM? (7)
 b) What is the “Kernel trick”? Explain different types of kernel functions. (7)

OR

- 16 a) Explain the pros and cons of backpropagation algorithm (6)
 b) Write short notes on any 4 activation functions used in Neural Networks (8)

Module IV

- 17 a) Suppose that the machine learning task is to cluster points representing into three clusters, where the points are A1(2, 10), A2(2, 5), A3(8, 4), B1(5, 8), B2(7, 5), B3(6, 4), C1(1, 2), C2(4, 9). The distance function is Euclidean distance. Suppose initially we assign A1, B1 and C1 as the center of each cluster, respectively. Use the k-means algorithm to show (7)
 a) The three cluster centers after the first round of execution
 b) The final three clusters

- b) Explain the methods used to learn multiple classes for a K class Classification problem. (7)
- 18 a) Find the covariance matrix of the given table (6)

Features	Sample 1	Sample 2	Sample 3	Sample 4
X1	4	8	13	7
X2	11	4	5	14

- b) State the PCA algorithm. Is principal component analysis a supervised learning problem? Justify your answer. (8)

Module V

- 19 a) A binary classification result, where the correct labels are [T, T, F, T, F, T, F, T] and the predicted labels are [T, F, T, T, F, F, F, T]. Assume T means “true” (the desired class) and F (“false”) is the “default” class. Compute accuracy, and compute recall, precision, and F1 for T. (6)
- b) Assume you have a model with a high bias and a low variance. What are the characteristics of such a model? (8)

OR

- 20 a) Use the single and complete link agglomerative clustering to group the data described by the following distance matrix. Show the dendrograms (8)

	A	B	C	D
A	0	1	4	5
B		0	2	6
C			0	3
D				0

- b) Assume the following: A database contains 80 records on a particular topic of which 55 are relevant to a certain investigation. A search was conducted on that topic and 50 records were retrieved. Of the 50 records retrieved, 40 were relevant. Construct the confusion matrix for the search and calculate the precision and recall scores for the search. (6)
